Artificial Intelligence Fundamentals

SEARCH LABORATORY (OPTIONAL)

Searching in a grid-world

Suggested environment: https://minigrid.farama.org/environments/minigrid/

- E.g., Crossing - MiniGrid Documentation
- *pip install minigrid* required in addition to gymnasium

Interaction with the environment:

```python
def run_crossing_env(steps=1000):

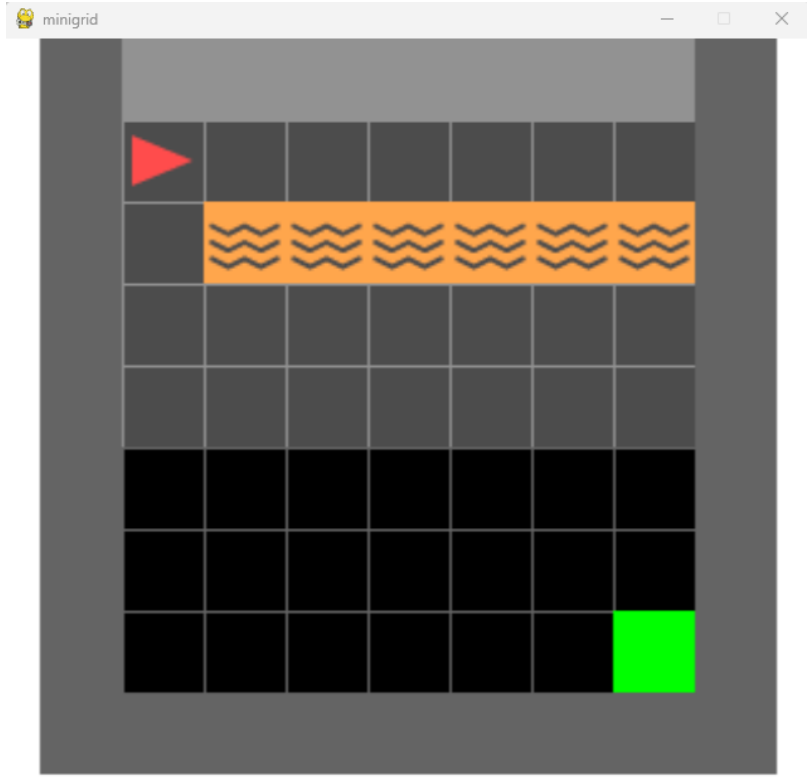
    # Crossing env doc
    # https://minigrid.farama.org/environments/minigrid/CrossingEnv/

    # discrete action space with 3 actions
    env = gymnasium.make("MiniGrid-LavaCrossingS9N1-v0", render_mode="human")

    observation, info = env.reset(seed=0)
    for _ in range(steps):
        env.render()
        action = env.action_space.sample()
        observation, reward, terminated, truncated, info = env.step(action)

        if terminated or truncated:
            break

    env.close()
```



avoid the lava and get to the green goal square

Searching in a grid-world

You can **implement** / **use**

- informed (A*)
- uninformed (DFS/BFS) search strategies

One library is https://pypi.org/project/pathfinding/

- https://github.com/brean/python-pathfinding/blob/main/notebooks/performance.ipynb
- Can you make gymnasium work with pathfinding?

Try out different grid world with different algorithms

Build easy/difficult/unsolvable grid worlds

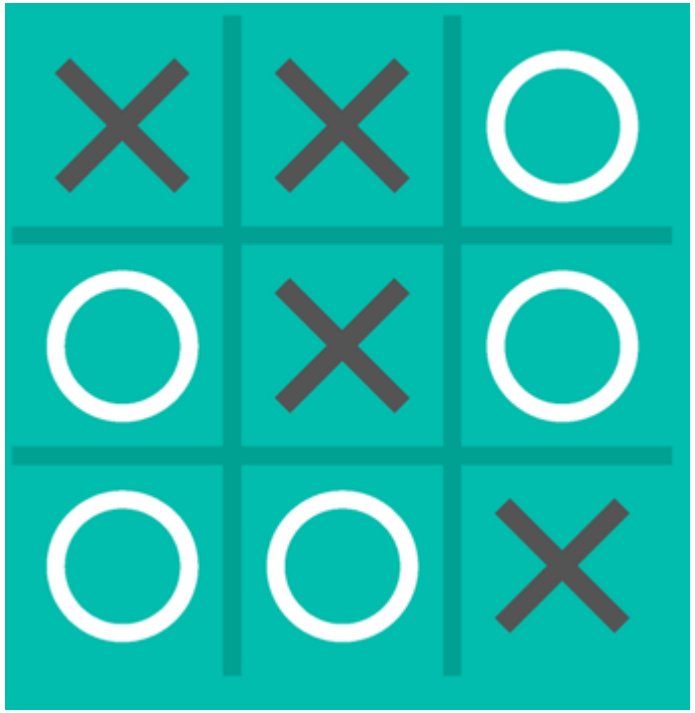
Scale to more complex environments

Make the environment non-deterministic?

Tic tac toe

Gym-like environment provided

Can you solve it with minimax / alpha-beta?



Generative AI for coding

Useful to quickly get used to a new library

Useful to integrate different libraries together

Useful to plot stuff / log results